

HOTEL BUSINESS DATA ANALYSIS

Investigating Booking & Cancellation Behaviour

A Data Visualization Project

119,390

Records Analyzed

29

Features

2017-2019

Period Covered

Tools: Python · Pandas · NumPy · Matplotlib · Seaborn · Jupyter Notebook

Report Date: August 2026

1. Executive Summary

This report analyzes 119,390 hotel bookings spanning 2017–2019 across two properties — a City Hotel and a Resort Hotel — to understand what drives guests to book and, critically, what drives them to cancel. After a rigorous data-cleaning process, 85,962 valid booking records were retained for analysis.

Three core business questions anchor the investigation:

- Which hotel type do customers book most often, and how does demand shift across the year?
- Does the length of stay affect the probability that a booking is cancelled?
- Does lead time — the gap between booking and arrival — affect the cancellation rate?

The analysis finds that the City Hotel is both more popular (61% of bookings) and more cancellation-prone (30.2% vs. 23.5% for the Resort Hotel). Cancellation risk rises sharply with both longer stays and longer booking lead times, and is heavily concentrated among guests who book through Online Travel Agencies, make no changes to their reservation, and travel for personal (non-business) reasons. These patterns translate directly into five actionable recommendations covering deposit policy, targeted reminders, dynamic pricing, and channel strategy, detailed in Section 7.

61.0% City Hotel Share	27.5%* Overall Cancel Rate	75.4% OTA/TA Dependency
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**Blended estimate across both hotel types on the cleaned dataset.*

2. Problem Statement & Objectives

2.1 Why Booking Behaviour Matters

Understanding customer booking behaviour is central to a hotel's commercial performance. Every cancelled booking represents lost or displaced revenue, wasted marketing spend to originally acquire that guest, and a room that may go unsold if it cancels too close to arrival to be resold. Conversely, understanding what drives demand — which hotel type, which season, which channel — lets a hotel allocate staff, pricing, and inventory ahead of time rather than reactively.

Concretely, insight into booking behaviour can improve decisions such as:

- Revenue management: setting room rates and minimum-stay rules by season and lead-time window.
- Overbooking strategy: calibrating how many extra rooms to sell based on expected cancellation rates for a given segment.
- Marketing & channel investment: deciding how much to spend on OTAs versus direct-booking incentives.
- Operations & staffing: anticipating occupancy peaks and troughs across the calendar.
- Guest communication: deciding who needs a reminder, deposit, or confirmation call to reduce no-shows.

2.2 Business Questions

- Q1 — Hotel type popularity: Which hotel type (City or Resort) do customers book most often, and how does demand vary month to month?
- Q2 — Stay duration: Does the total length of stay influence the likelihood that a booking is later cancelled?
- Q3 — Lead time: Does the gap between the booking date and the arrival date (lead time) influence the cancellation rate?

2.3 Project Objective

The single main deliverable is a cleaned, analysis-ready dataset paired with a set of clear, business-facing visualizations that answer the three questions above — culminating in a concise set of data-backed recommendations a hotel manager can act on without needing to read any code.

3. Data Overview

The raw dataset contains 119,390 bookings and 29 columns, covering arrivals from 2017 through 2019 across two hotel properties — a City Hotel and a Resort Hotel.

3.1 Columns Most Relevant to the Business Questions

Column	Description
hotel	Property type: City Hotel or Resort Hotel
is_canceled	Target variable — 1 if the booking was cancelled, 0 otherwise
lead_time	Days between the booking date and the arrival date
arrival_date_month / year	Calendar timing of the stay
stays_in_weekend_nights / weekdays_nights	Combined to derive total_stay (length of stay)
adr	Average Daily Rate — the price paid per night
customer_type	Personal, Family, Contract, or Business booking
market_segment	Booking channel, e.g. Online TA, Direct, Corporate
booking_changes	Number of modifications made to the reservation
meal	Meal package selected (Breakfast, Half Board, Full Board, None)

4. Data Preprocessing

Before any analysis, the raw dataset was assessed and cleaned. Every decision below was made to preserve as much genuine signal as possible while removing rows or values that would distort the business questions.

4.1 Missing Values

Column	Missing Value Meaning	Treatment
children	Guest simply travelled with no children	Filled with 0
city	No city was captured for the guest	Filled with 'Unknown'
agent	Booking was not placed through a travel agent	Filled with 0
company	Booking was not made on behalf of a company	Filled with 0

In each case, a missing value does not mean 'unknown data' so much as 'this attribute does not apply to this guest' — for example, a blank agent ID most plausibly means the guest booked directly rather than through an intermediary. Filling with 0 preserves that guests differ meaningfully from those with a real agent/company ID, without discarding the row.

4.2 Duplicate Rows

Exact duplicate rows were identified and removed using `drop_duplicates()`. Retaining duplicates would have double-counted the same booking event and inflated both demand and cancellation statistics.

4.3 Inconsistent / Unclear Values

The meal column contained an 'Undefined' category, which — per the hotel's own booking system — is used interchangeably with 'no meal package selected.' These rows were recategorized as 'No Meal' so the meal-preference analysis reflects genuine guest choice rather than a data-entry artifact.

4.4 Anomalies

Two anomaly types were identified and filtered out:

- Invalid ADR: rows with a negative average daily rate (impossible) or an extreme outlier value ($\geq$ €5,000/night, implausible for these two properties) were removed.
- Zero-occupancy bookings: rows where adults + children + babies summed to zero — meaning no guest was actually recorded — were removed, since these cannot represent a real stay.

4.5 Feature Engineering

A `total_stay` feature was engineered by summing `stays_in_weekend_nights` and `stays_in_weekdays_nights`, giving a single length-of-stay metric used throughout Section 5.2.

85,962

Rows Remaining After Cleaning

5. Data Analysis

With a clean dataset in hand, the three business questions were answered in turn, each backed by dedicated visualizations.

5.1 Hotel Type Popularity & Monthly Demand

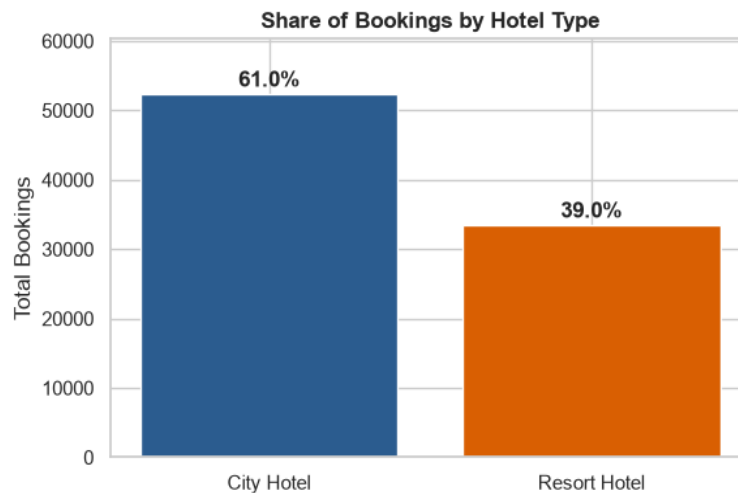


Figure 1. Share of total bookings by hotel type.

The City Hotel is booked considerably more often than the Resort Hotel — 61.0% of all bookings versus 39.0%. This is a meaningful gap: for every two City Hotel bookings, the Resort Hotel receives roughly one.

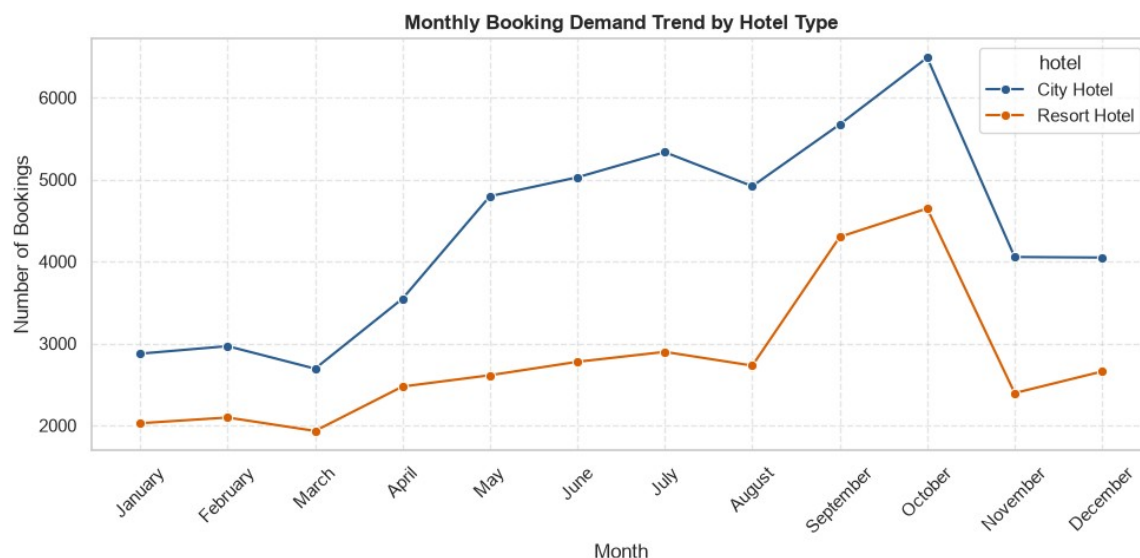


Figure 2. Number of bookings per month, by hotel type (2017–2019 combined).

Demand for both hotel types is lowest in the winter months (January–March) and rises through spring into a peak in the autumn. The City Hotel peaks sharply in October (~6,500 bookings) after climbing steadily from April onward, then drops abruptly in November before flattening. The Resort Hotel follows a gentler, more classically seasonal curve, rising through summer and peaking in September–October (~4,300–4,650 bookings) before tapering off.

This pattern is consistent with two overlapping demand drivers. The City Hotel's steep spring-to-autumn climb, and especially its October peak, points to business travel, conferences, and city-break tourism concentrated in the shoulder season. The Resort Hotel's smoother summer build is more consistent with classic leisure and holiday travel, where families and couples book warm-weather getaways.

Operationally, this seasonality supports several decisions: staffing and inventory can be scaled up ahead of the September–October peak for both properties; the City Hotel's sharp November drop-off is a natural window for renovation, maintenance, or promotional discounting to smooth demand; and marketing spend for the Resort Hotel is best front-loaded in spring to capture summer bookings before the peak.

5.2 Impact of Stay Duration on Cancellation Rate

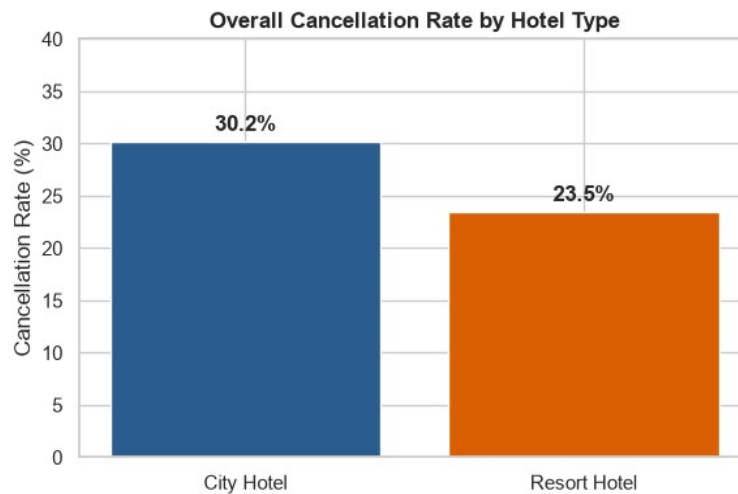


Figure 3. Overall cancellation rate by hotel type.

The City Hotel is cancelled far more often than the Resort Hotel — 30.2% versus 23.5% of bookings. Combined with the City Hotel's larger booking volume, this means the majority of cancelled room-nights in the business originate from the City property.

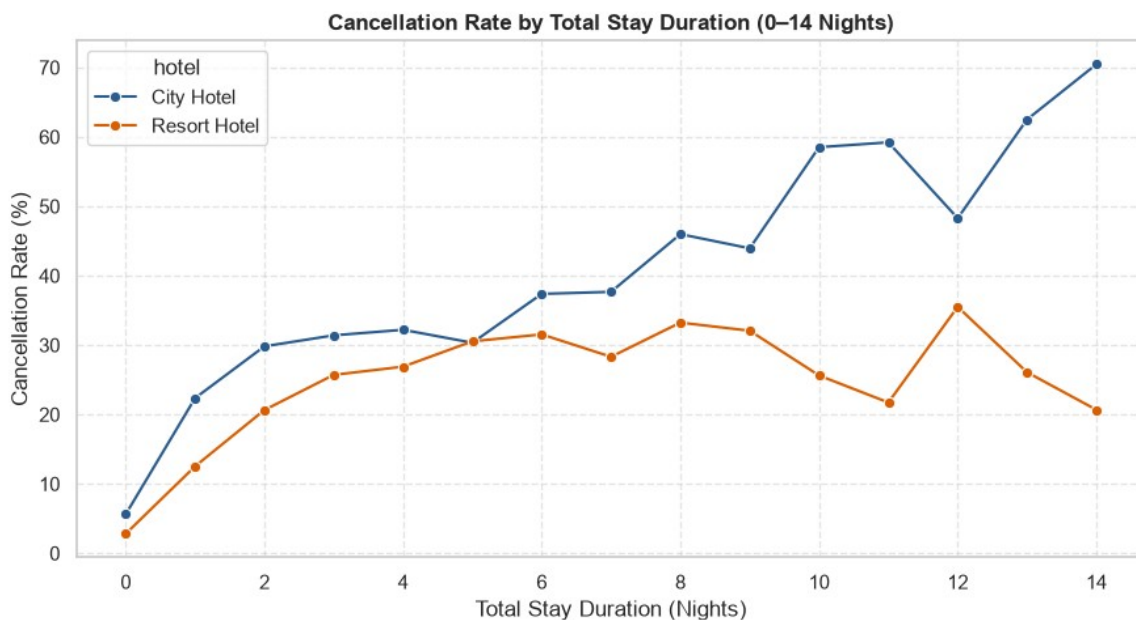


Figure 4. Cancellation rate by total length of stay, 0–14 nights, by hotel type.

The relationship between stay duration and cancellation is clearly positive for both hotel types, but far stronger — and far noisier — for the City Hotel. City Hotel cancellations climb from roughly 5% for a single-night stay to over 70% for a 14-night stay, a near linear increase with some volatility at the longer end. The Resort Hotel shows the same upward direction but much more moderately, rising from around 3% at one night to a plateau in the 25–35% range for stays of 5–13 nights, before dipping again at 14 nights.

In short: the relationship is positive at both properties, but it is steeper and more erratic at the City Hotel — longer City Hotel stays are disproportionately likely to fall through.

Two plausible explanations for why longer stays cancel more often:

- Planning uncertainty compounds over time. A longer stay is booked further into an uncertain future (work schedules, other travel plans, budget changes), so there is simply more time and more reasons for circumstances to change before arrival.
- Longer, pricier bookings are more often used as 'placeholders.' Guests — especially business travellers or those booking through OTAs with free cancellation — may reserve a long block of nights speculatively and cancel or shorten it once their actual plans firm up, a behaviour more common in a City Hotel's business-and-events-driven customer base than a Resort Hotel's leisure guests.

5.3 Impact of Lead Time on Cancellation Rate

Lead time is the number of days between the date a booking is made and the guest's arrival date — in other words, how far in advance the reservation is placed.

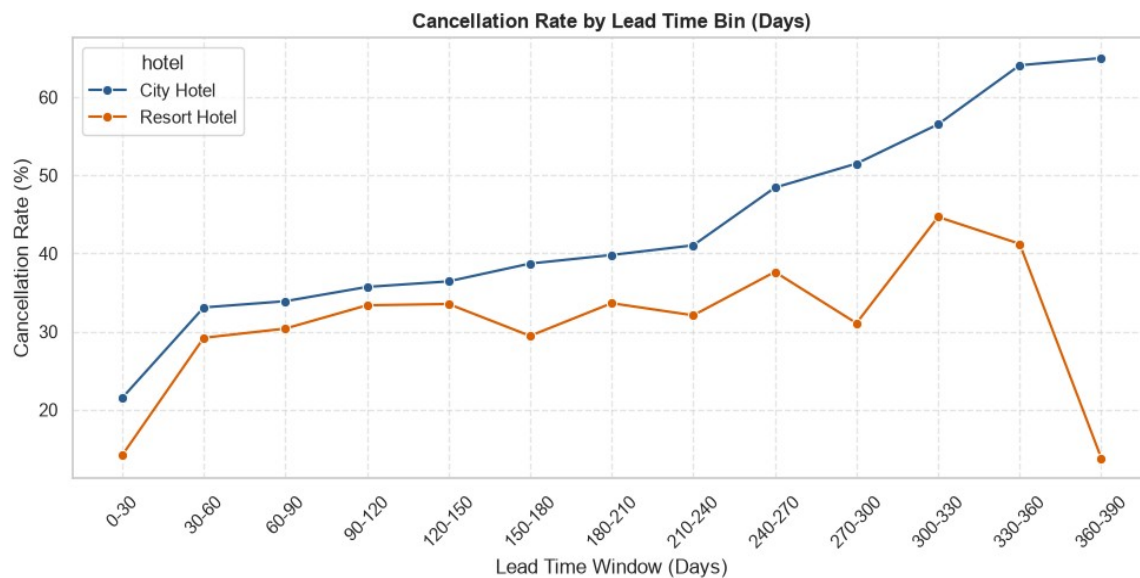


Figure 5. Cancellation rate by lead-time window (30-day bins), by hotel type.

Cancellation is lowest at very short lead times for both hotel types: bookings made 0–30 days before arrival cancel at only 21.6% (City) and 14.2% (Resort) — these are guests who know their plans are firm because the trip is imminent. From there, cancellation risk climbs steadily as lead time increases. The City Hotel rises almost monotonically, reaching roughly 65% cancellation for bookings made 360+ days in advance — the highest

cancellation rate of any segment in the dataset. The Resort Hotel is comparatively more stable in the mid-range (roughly 30–38% across 60–270 days) before spiking to around 41–45% in the 300–360 day window.

The two hotel types diverge clearly at long lead times: the City Hotel's cancellation rate keeps climbing with no ceiling in view, while the Resort Hotel stays comparatively flatter through the middle of the range and only rises meaningfully in the final few bins. A plausible explanation is that City Hotel bookings that far in advance are disproportionately speculative business or event bookings — placeholders subject to change — whereas Resort bookings, tied to planned holiday travel, tend to be made with firmer intent even well ahead of time.

6. Additional Insights

Beyond the three core questions, several supplementary cuts of the data sharpen the recommendations in Section 7.

6.1 Cancellation Risk by Customer Type

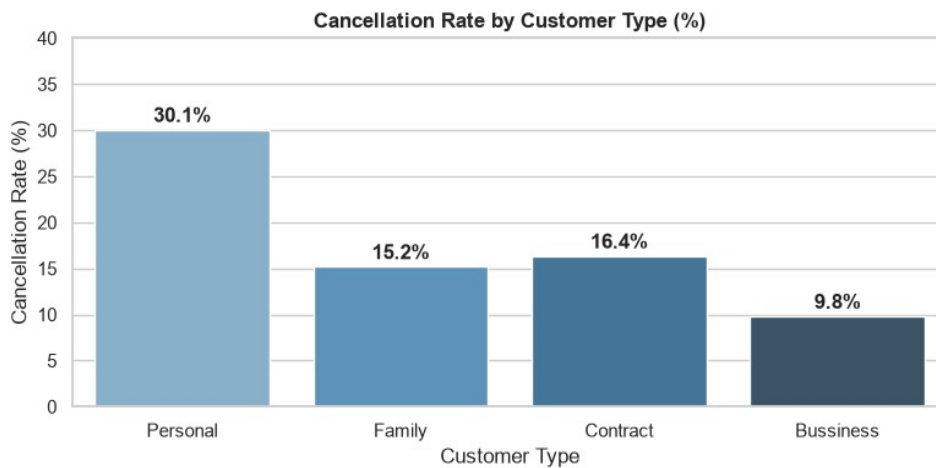


Figure 6. Cancellation rate by customer type.

Personal / Transient bookings cancel far more often (30.1%) than any other segment — roughly 3x the rate of Business bookings (9.8%), and nearly double Contract (16.4%) and Family (15.2%) bookings. Business and Contract guests, who typically book under firmer institutional commitments, are the business's most reliable segments.

6.2 Cancellation and Booking Modifications

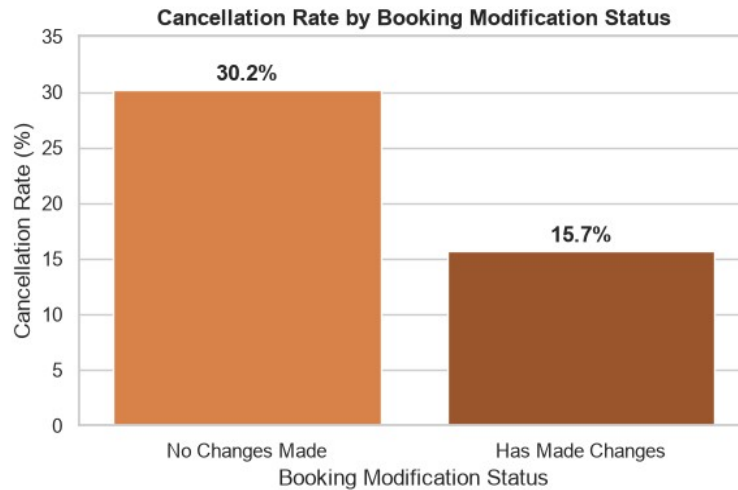


Figure 7. Cancellation rate by whether the guest modified their booking.

Guests who make at least one change to their reservation (date, room type, etc.) cancel at roughly half the rate of those who never touch their booking — 15.7% versus 30.2%. A modification is a strong positive signal of continued engagement and intent to stay.

6.3 Seasonal Pricing (ADR)



Figure 8. Average Daily Rate (€) by month and hotel type.

Resort Hotel pricing is highly seasonal, climbing from roughly €48–58 in the low season (Jan–May) to a peak above €185 in the autumn high season — more than triple. City Hotel pricing is comparatively stable year-round, moving in a much narrower band of roughly €85–130, consistent with steadier, less leisure-driven demand.

6.4 Booking Channel Dependency

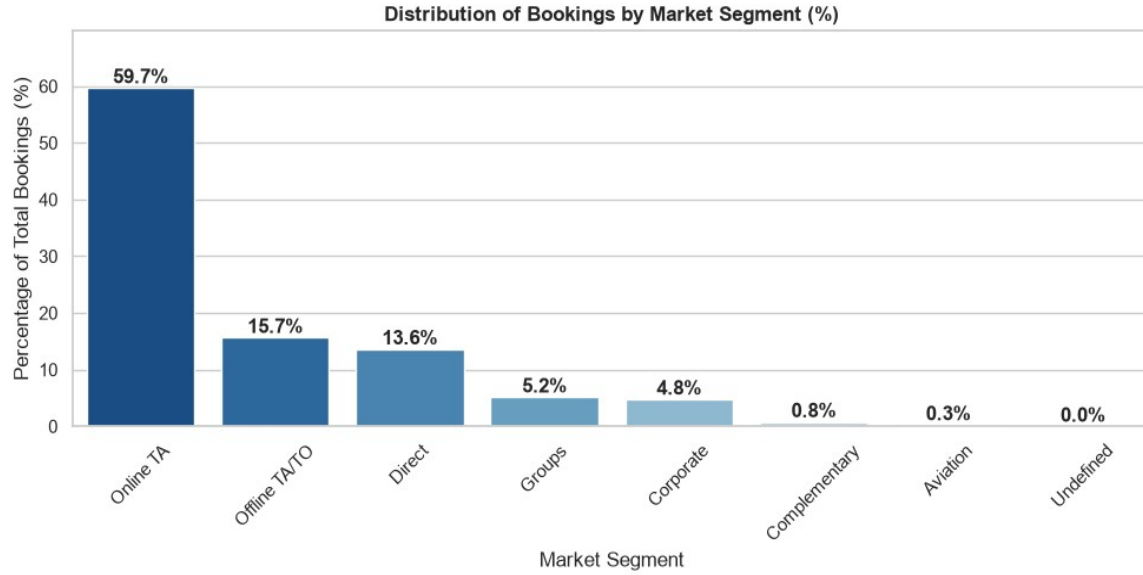


Figure 9. Distribution of bookings by market segment.

Online Travel Agencies dominate booking volume at 59.7% of all bookings, and combined with Offline TA/TO (15.7%), third-party intermediaries account for 75.4% of total bookings. Only 13.6% of guests book Direct. This heavy OTA dependency has real commission-cost implications and represents an opportunity for direct-booking incentives.

6.5 Meal Package Preference

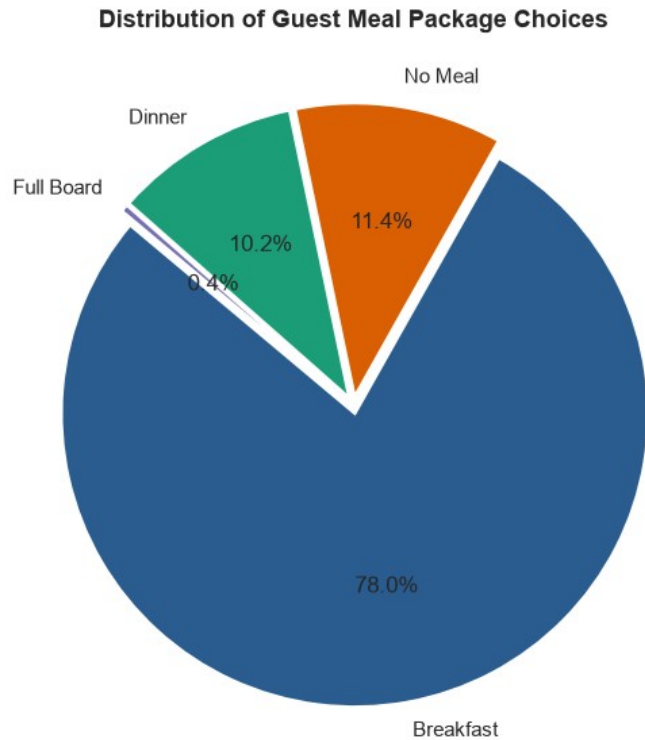


Figure 10. Distribution of guest meal package choices.

The large majority of guests (78.0%) select Bed & Breakfast. Half Board / Dinner is a distant second at 10.2%, no meal plan at all accounts for 11.4%, and Full Board is negligible (0.4%) — suggesting limited demand to expand full-board packaging.

7. Summary of Key Findings

- Hotel type & seasonality: The City Hotel drives the majority of volume (61%) but also the majority of cancellations (30.2% vs 23.5%). Demand for both properties peaks in September–October and troughs in winter, with the City Hotel showing a sharper, more business-travel-like curve and the Resort Hotel a smoother, leisure-like curve.
- Stay duration: Cancellation risk rises steadily with length of stay at both properties, but far more steeply at the City Hotel (up to ~70% for 14-night stays) than the Resort Hotel (plateauing around 25–35%).
- Lead time: Cancellation risk is lowest for imminent bookings and climbs the further out a booking is made — most dramatically for the City Hotel, which reaches ~65% cancellation for bookings made a year or more in advance.
- Underlying drivers: Personal/Transient guests, bookings with no modifications, and OTA-sourced bookings are consistently the highest-risk segments across all three cuts.

8. Business Recommendations

8.1 Hotel Type & Seasonality

- Grow the Resort Hotel's off-peak volume with targeted spring promotions (Jan–May currently just €48–58 ADR) to smooth its heavily skewed seasonal curve and reduce dependence on the September–October peak.
- Capitalize on the shared September–October peak by locking in staffing and inventory early, and by testing modest early-bird rate increases at both properties, since demand at that time is proven and resilient.

8.2 Stay Duration & Cancellations

- Introduce a graduated deposit or partial-prepayment policy for bookings of 5+ nights at the City Hotel in particular, where cancellation risk rises fastest with stay length — a deposit converts a 'placeholder' booking into a committed one.
- Offer a modest rate discount for guests who commit to a minimum-stay, non-refundable or partially-refundable rate on longer bookings, trading a small price concession for a large reduction in cancellation risk.

8.3 Lead Time & Far-Ahead Bookings

- Automate a confirmation/reminder sequence (e.g., at 90, 30, and 7 days before arrival) specifically for bookings made 180+ days in advance, where cancellation risk is highest — a well-timed nudge can convert an uncertain plan into a firm one.

- Require a small deposit for bookings placed more than 6 months ahead — the segment with a ~65% cancellation rate at the City Hotel — while keeping late, near-term bookings deposit-free to avoid friction where cancellation risk is already low.

8.4 Channel & Segment Strategy

- Invest in direct-booking incentives (loyalty pricing, free breakfast upgrades) to shift share away from the 75.4%-dependent OTA/TA channel, which carries both commission cost and, empirically, higher cancellation rates than direct bookings.
- Prioritize retention offers for Business and Contract customer types, the two segments with the lowest cancellation rates (9.8% and 16.4%), and treat Personal/Transient bookings — the highest-risk segment at 30.1% — as the primary target for deposit and reminder policies.

8.5 Highest-Impact Recommendation

Of the above, the deposit/prepayment policy for far-ahead City Hotel bookings (8.3) is likely to have the single biggest impact. The evidence for this is threefold: (1) the City Hotel carries the largest booking volume, so any rate reduction there has outsized revenue effect; (2) its cancellation rate for 360+ day lead times, at ~65%, is the single highest risk segment identified anywhere in the data (Figure 5); and (3) the booking-modification analysis (Figure 7) shows that any mechanism which increases guest commitment — a deposit functions the same way — roughly halves cancellation likelihood. Together, these point to far-ahead City Hotel bookings as the highest-leverage place to intervene.

9. Conclusion

This analysis confirms that cancellation behaviour at both hotels is not random — it follows clear, actionable patterns tied to hotel type, stay length, lead time, customer type, booking channel, and guest engagement (modifications). The City Hotel, while the larger revenue driver, is also the higher-risk property across every dimension examined. Implementing lead-time-based deposits, stay-length incentives, and a shift toward direct bookings and higher-commitment customer segments offers a realistic, data-backed path to reducing the revenue currently lost to cancellations.

Prepared as part of the Investigate Hotel Business using Data Visualization project · Python · Pandas · Matplotlib · Seaborn